

LAST NAME: _____

Practice Quiz 03B — MATH 2055 — Fall 2026

Practice for Quiz 03

The following is closed book and closed note. Don't assume the grader will know what you mean. Use full sentences and provide complete and correct solutions.

Revised using GPT-6 Astra in the Codex harness.

1. Determine whether each statement is true or false. For a true statement, give a formula for y in terms of x . For a false statement, give a specific x for which no allowed y works and explain why.

(a) $\forall x \in \mathbf{R} \exists y \in \mathbf{R} (2y = x)$.

(b) $\forall x \in \mathbf{Z} \exists y \in \mathbf{Z} (2y = x)$.